

# Revolutionizing Culturally Relevant Math Education Through AI and Co-Designing Strategies

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## Abstract

There is a problem with student math proficiency. In 2023, Baltimore City Public Schools (BCPS) test results revealed a critical math proficiency gap, with none of 1,736 students from 13 schools achieving proficiency on the Maryland state math exam. This issue extends statewide, with less than a quarter of Maryland K-12 students proficient in math. To address this, we propose MathWiz, an intelligent tutoring system integrating culturally responsive teaching, mastery learning, and AI to revolutionize math education. Featuring a 3D avatar math companion, MathWiz provides personalized, adaptive instruction tailored to students' needs and cultural contexts. The project hypothesizes that integrating culturally relevant content with mastery learning will enhance math proficiency, engagement, and attitudes. Expected outcomes include improved math skills, critical thinking, and digital literacy, fostering equitable learning through scalable, inclusive solutions in underserved communities.

## CCS Concepts

• **Human-centered computing** → **Web-based interaction**; • **Applied computing** → **Interactive learning environments**; • **Computing methodologies** → **Intelligent agents**.

## Keywords

Intelligent Tutoring System, Mastery Learning, Mathematics, Culturally Relevant Learning, Adaptive Learning System

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## 1 Introduction

Mathematics proficiency is a critical determinant of academic success and future career opportunities, with Algebra even being called a gatekeeper to success in education and career endeavors. In Baltimore, Maryland, the state of math literacy is alarmingly low. Recent reports reveal that in 2023, 40% of Baltimore City high schools had zero students testing proficient in math [21]. This crisis is not confined to Baltimore; statewide, math literacy rates are concerning, with less than a quarter of Maryland K-12 students achieving proficiency in math [12]. Nationally, the situation is similarly troubling. According to the National Assessment of Educational Progress (NAEP), only 26% of 12th-grade students scored at or above the proficient level in mathematics [7].

To address challenges in mathematics education, innovative tools like MathWiz are being developed to improve learning outcomes. MathWiz is an intelligent tutoring system (ITS) that uses artificial intelligence (AI) to provide personalized, adaptive learning experiences tailored to individual student needs. Studies have shown that ITSs can significantly enhance engagement and comprehension in mathematics, particularly through immediate feedback and targeted support [1, 26]. By allowing students to learn at their own pace, these systems foster a deeper understanding of mathematical concepts and help bridge gaps in traditional education methods. MathWiz leverages these strengths to empower students and promote success in mathematics.

MathWiz not only personalizes learning but also incorporates culturally responsive teaching practices to make math more relatable to students. Research suggests that integrating cultural relevance into education can improve student engagement and achievement [9, 16]. To support this, MathWiz includes examples and narratives that reflect diverse backgrounds, helping students see connections between math and their own experiences. By combining AI-driven personalization with culturally informed content, MathWiz aims to create a more inclusive learning environment that supports both academic understanding and student engagement.

## 2 Related Work

### 2.1 Intelligent Tutoring Systems

Intelligent Tutoring Systems have revolutionized mathematics education by leveraging artificial intelligence to deliver personalized and adaptive learning experiences. Systems such as ActiveMath, Lexue100, AnimalWatch, Wayang Outpost, ALEKS, and Carnegie Learning’s Cognitive Tutor have contributed significant advancements in this field, each bringing unique methodologies that serve as a foundation for MathWiz. These systems demonstrate how AI can enhance learning by addressing gaps in student understanding and improving engagement. MathWiz synthesizes these strengths while addressing limitations to create a more inclusive, culturally responsive, and dynamic learning platform.

ActiveMath supports exploratory learning through personalized content and external tools like computer algebra systems, fostering cognitive growth [18]. Its emphasis on cognitive customization helps students deepen their understanding of complex problems, making it a powerful tool for learning. However, ActiveMath overlooks the cultural and emotional dimensions of education, which are critical for fostering a well-rounded learning experience. MathWiz addresses this gap by embedding culturally responsive teaching and mastery learning principles, while also introducing a 3D avatar companion. This approach not only enhances cognitive learning but also promotes inclusivity, helping students connect mathematical concepts to their lived experiences.

Lexue100 employs big data analysis to deliver adaptive feedback and incremental mastery through curriculum-aligned quizzes [27]. Its user-friendly design, gamification features, and instant feedback sustain student engagement, making it an effective tool for improving learning outcomes. However, the platform’s lack of real-time interactivity, such as conversational support, limits its ability to address immediate student needs. MathWiz builds on these strengths by integrating a conversational avatar that provides instant guidance and fosters active engagement, ensuring students feel supported and motivated throughout the learning process.

AnimalWatch focuses on improving arithmetic and fraction skills for middle school students through adaptive problem selection and scaffolding [5]. While effective for building foundational math skills, its narrow focus and limited session durations restrict its scalability and broader learning impact. Furthermore, its design does not address the need for cultural relevance or long-term academic growth. MathWiz expands on AnimalWatch’s framework by incorporating mastery learning principles, culturally responsive content, and personalized experiences tailored to diverse learners. By addressing these broader challenges, MathWiz ensures a more inclusive and meaningful educational experience that fosters both skill development and long-term success.

Wayang Outpost provides a unique approach to preparing students for the Math section of the SAT through narrative-based challenges and animated STEM role models [2]. Its multimedia animations, alternative teaching strategies, and problem-solving adventures offer a dynamic and engaging learning experience. The inclusion of female animated characters as STEM role models is particularly impactful, addressing the underrepresentation of women in math-related fields. MathWiz builds on Wayang Outpost’s strengths

by enhancing storytelling components and integrating culturally diverse role models, further promoting equity and inspiring students from all backgrounds.

ALEKS is an adaptive platform that evaluates students’ knowledge and provides personalized lessons to help them master concepts at their own pace [22]. Its features include personalized assessments, problem-solving exercises, and dashboards that highlight areas for improvement. However, ALEKS often struggles to maintain student engagement due to its limited adaptability to diverse learning preferences and lack of interactivity. MathWiz addresses these challenges by tailoring experiences to visual, auditory, and kinesthetic learners while incorporating diverse role models and accessible design features. This approach ensures inclusion and fosters a deeper appreciation for mathematics across all students.

Carnegie Learning’s Cognitive Tutor (CT) features interactive lessons, review modes, and handwriting recognition for equation-solving [23]. It offers a “pencil-and-paper” experience that logs data to analyze learning patterns, providing valuable insights into student progress. However, the platform faces challenges such as simplified skill tracking, error-prone handwriting recognition, and scalability issues. MathWiz builds on CT’s strengths by improving tracking accuracy, expanding personalization, and ensuring scalability to reach underserved communities. These enhancements make MathWiz a more robust and equitable tool for math education.

By synthesizing the strengths of these ITSs, personalization, adaptivity, and innovative problem-solving tools, MathWiz takes an innovative approach to addressing educational inequities. Its integration of culturally responsive content, mastery learning, and co-design methodologies positions it as a unique tool for creating equitable, engaging, and meaningful learning experiences. MathWiz not only aims to improve math skills but also to inspire a deeper connection with the subject, ensuring all students have the opportunity to succeed.

### 2.2 Embodied Conversational Agents

Embodied Conversational Agents (ECAs) enhance learning by simulating human interactions through verbal and non-verbal cues, which foster engagement, motivation, and adaptive instruction [6, 15, 17]. They have been widely used in various domains, including education, to enhance user engagement and learning outcomes. By providing personalized feedback, emotional support, and real-time guidance, ECAs foster interactive and supportive learning environments. Research indicates that ECAs effectively personalize learning experiences, improving satisfaction and reducing cognitive load by adapting to learners’ needs [6, 14]. In mathematics education, ECAs have proven effective in reducing math anxiety and building student confidence through real-time, personalized feedback [15]. By integrating students’ cultural and linguistic contexts, ECAs also support culturally responsive pedagogy, making learning more relatable and reducing barriers to engagement [4, 16, 19, 20].

A key strength of ECAs lies in digital identity representation, which helps students feel seen and valued, fostering a stronger sense of belonging, enhancing engagement, and boosting self-efficacy. Studies show that students connect more with avatars that reflect their cultural identity, language, or traits, increasing motivation

and inclusivity [13]. Customizable avatars also help mitigate biases and promote equity [17]. Building on these advantages, MathWiz integrates an avatar-based 3D ECA with AI-driven dialogue systems [10] to provide cognitive support and culturally responsive interactions.

### 2.3 Culturally Responsive Learning

Culturally responsive learning is an educational approach that integrates students' cultural backgrounds, experiences, and identities into the learning process. By making learning experiences more relatable, educators can foster greater engagement and motivation among students, particularly those from diverse or underrepresented groups [11]. This approach helps bridge the gap between students' lived experiences and the content they are learning, enhancing both academic achievement and emotional well-being. Culturally responsive learning emphasizes the importance of recognizing and validating students' identities, creating an inclusive and supportive classroom environment.

In a study, the authors discuss how culturally responsive teaching practices can improve student engagement and outcomes in mathematics education. The study emphasizes that traditional math instruction often overlooks students' cultural contexts, leading to disengagement and inequitable learning experiences. By incorporating students' cultural identities and real-world contexts, educators can provide more meaningful learning opportunities [3]. For MathWiz, incorporating culturally responsive learning practices, such as ensuring that the 3D avatar and the adaptive content reflect and validate students' identities, would not only enhance engagement but also foster critical thinking. This approach aligns with MathWiz's mission to address systemic inequities in education by providing a more inclusive and culturally responsive learning environment.

### 2.4 Mastery Learning

Mastery learning is an instructional approach in which students must demonstrate proficiency in a topic before moving on to more advanced material. This method emphasizes individualized learning, allowing students to progress at their own pace until they achieve a certain level of mastery. The approach contrasts with traditional models, where students are expected to progress through content at the same pace, regardless of their understanding [8]. Mastery learning has been shown to enhance student outcomes by providing a clear path to success, encouraging mastery of fundamental concepts before advancing, and reducing the gap in performance between students. It is particularly beneficial in environments that aim to address diverse learning needs and ensure all students have the opportunity to succeed. A study discusses the impact of mastery learning on intelligent tutoring systems (ITS) and highlights how integrating this model into personalized education environments can improve learning outcomes [24]. By ensuring students master foundational skills before progressing, mastery learning fosters deeper understanding, long-term retention, and academic success. ITS enhance this approach by adapting to individual needs, offering tailored feedback and support throughout the learning process. This combination reduces performance gaps between students and creates opportunities for equitable, scalable education, particularly

benefiting diverse or underrepresented learners. The study underscores the importance of designing educational technologies that prioritize individualized learning and foundational skill mastery.

## 3 System Overview

Noting that MathWiz is still in development, the current system consists of four core components: (i) a personalization module that collects user data and uses it to build a digital model representation of the user, (ii) an adaptive learning engine that evaluates proficiency and customizes learning paths, (iii) a dynamic lesson generation engine that creates culturally relevant and skill-appropriate lessons, and (iv) a 3D avatar math companion that serves as an interactive tutor, providing real-time guidance and fostering engagement. Each component integrates seamlessly with the system's frontend features, combining user profile creation, assessment tools, and an interactive dashboard, to deliver a comprehensive learning experience.

### 3.1 Personalization Module

To better understand and provide students with a pleasant and tailored educational experience, MathWiz builds a digital model representation of each student in the form of data. Throughout the application, the user's representation constantly changes as the system gains more information about the user. This dynamic model incorporates details such as learning preferences, progress in specific algebraic concepts, and cultural interests to create a highly personalized learning path. By continuously adapting, the system ensures that the educational content remains relevant, engaging, and suited to the user's unique needs. This approach not only enhances the learning experience but also fosters a deeper connection between the user and the material, increasing the likelihood of long-term success. Building a system centered around the user is essential, as it acknowledges individual learning differences, making education more effective and enjoyable. However, we recognize the ethical concerns surrounding data collection, privacy, and bias. MathWiz is committed to maintaining strict privacy safeguards, ensuring data is protected and never misused. Additionally, parents have full control over their child's data, including the ability to delete accounts upon request, reinforcing our dedication to transparency, ethical AI practices, and user privacy.

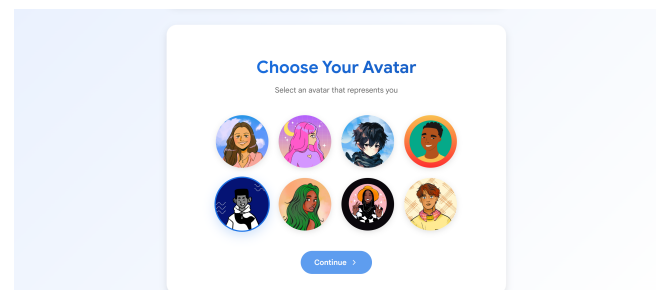


Figure 1: Profile Image Selection Page

**3.1.1 User Profile Creation.** When users first interact with MathWiz, after registering, users will undergo an onboarding flow in

which they pick their application preferences which include: a profile image that best represents them (Fig. 1), a choice between two math companions that will accompany them throughout their learning journey (Fig. 2), and a choice of general non-academic interests. All of these data points contribute to user modeling, inform content customization, and are used to create lessons that are tailored to the user's culture.

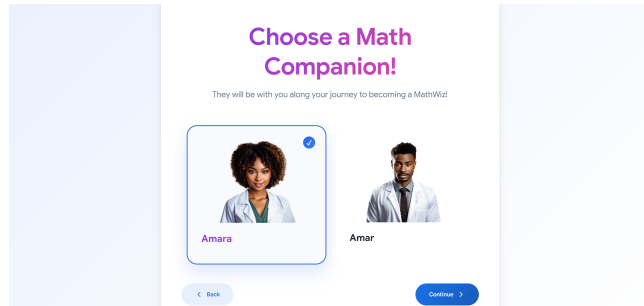


Figure 2: Avatar Selection Page

**3.1.2 3D Avatar Math Companion.** Throughout the student's time on MathWiz, they will always have a 3D avatar companion by their side, to tutor, teach, instruct, or answer questions. Currently, the system offers two selectable avatar companions, as shown in Fig. 2, Amara (to the left) and Amar (to the right), however, more avatars will be added in the future. These avatars have the ability to naturally understand, listen, and respond to the students appropriately, just as a typical tutor or teacher is able to. The avatars aim to reflect the diverse cultural identities in BCPS with 70% being African-American and 20% being Hispanic/Latino [25].

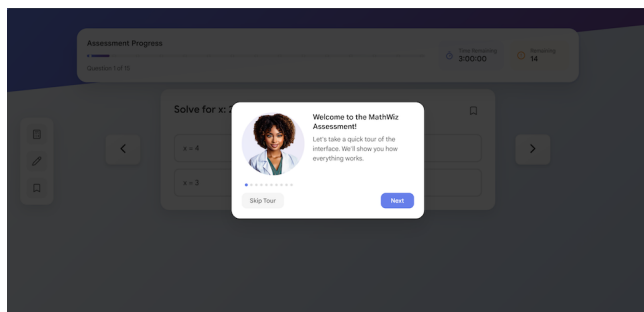


Figure 3: Avatar giving the user a tour of the assessment

The avatars also allow multimodal interaction, combining written text and speech to create a natural, engaging, and supportive learning environment. Some examples of the avatar's role in the system are displayed in the images below. In regards to the pre-assessment that students take, the 3D avatar interacts with the student and gives them a tour of the assessment interface (Fig. 3). Another example of the avatar's involvement is the lesson page. As shown in Fig. 4, the chosen 3D avatar is the primary instructor and delivers lesson content to the user.

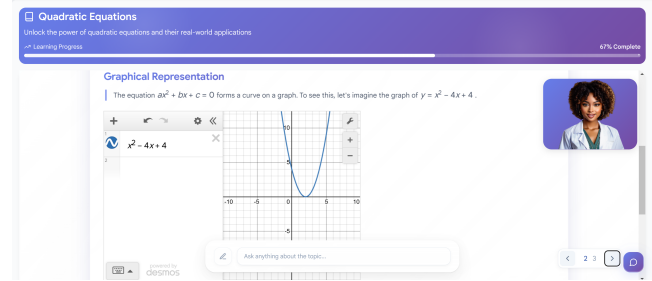


Figure 4: Snippet of avatar instructing during a lesson

## 3.2 Adaptive Learning Engine

MathWiz constantly monitors the user's progress in various mathematical concepts, allowing the system to move at the student's pace. Using a concept graph that maps relationships between algebraic topics, the system identifies gaps in prerequisite knowledge and adjusts lesson plans accordingly. This ensures that students achieve mastery in foundational concepts before advancing to more complex topics. Metrics such as concept accuracy, adjusted accuracy, and weighted accuracy are tracked to provide a comprehensive understanding of the student's strengths and areas for improvement. By adapting in real time, MathWiz creates a supportive learning environment that fosters confidence and long-term success in mathematics.

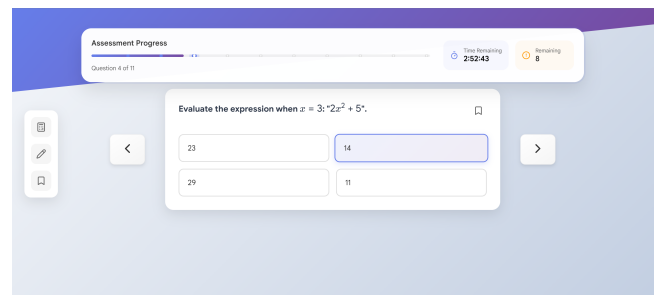
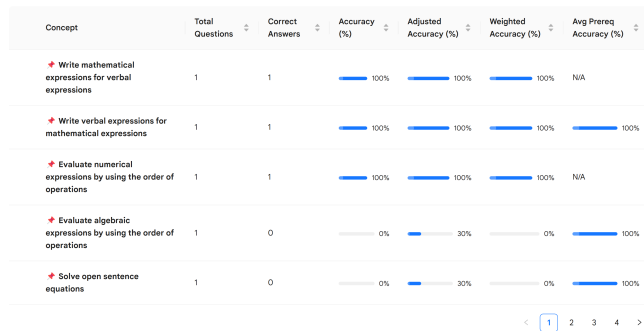


Figure 5: Pre-test Assessment

**3.2.1 Pretest Assessment.** After profile creation, students are directed to take a pre-test assessment to gauge the user's current algebra proficiency and place them on an appropriate learning path (Fig. 5). Students are provided with tools such as a calculator, scratchpad, and question bookmarking tool. The assessment covers major algebra concepts such as writing mathematical expressions for verbal expressions, solving open sentence equations, and more to get a good baseline of the student's proficiency in the various concepts. Each question on the assessment is linked to a concept, organized within MathWiz's concept graph. This graph represents concepts as nodes and links different nodes based on the relationship between them, the main one being prerequisites. After the pre-test is completed, the results are evaluated and a concept mastery map is created. As students learn and progress, their concept mastery map will be updated.



**Figure 6: Sample User Concept Mastery Distribution**

**3.2.2 Mastery Learning.** As shown in Fig. 6, in MathWiz’s implementation of mastery learning, each concept for a user has an accuracy, adjusted accuracy, weighted accuracy, and an average prerequisite accuracy. For each concept, the system tracks the total number of questions the user has been asked with the concept, the total number of correct answers, and the weighted consideration of each question’s difficulty. Concept accuracy is defined as the percentage of correct answers out of the total questions asked. The adjusted accuracy of a concept combines the accuracy of the direct concept with the accuracy of all prerequisites to give a greater contextual understanding of the student’s mastery of the concept. Currently, in this calculation, the accuracies of the concept and prerequisite are weighted (70% concept, 30% prerequisites) to reflect their relative importance. The adjustment factor is important because a student may perform poorly on a concept not because of a lack of understanding of that concept itself, but due to gaps in prerequisite knowledge. By considering prerequisite mastery, the system provides an accurate representation of the student’s true understanding and learning needs. Weighted accuracy adjusts the regular accuracy of a concept depending on the difficulty level of the questions. A more difficult question will carry more weight, than a question with lower difficulty. Lastly, the average prerequisite accuracy calculates the average mastery levels for all prerequisites for a concept, excluding the concept itself in the calculation.

### 3.3 Dynamic Lesson Generation Engine

**3.3.1 Pedagogical Framework.** All lessons in MathWiz follow a pedagogical framework with specific general rules. These include assessing prior student knowledge and balancing conceptual understanding with procedural fluency. Additionally, lessons focus on checking for understanding throughout the learning session and employing an efficient, standard problem-solving strategy. This framework aims to facilitate and support productive struggle. Each lesson is divided into 6 sections, called pedagogy steps: introduction, concept explanation, example problems, practice problems, summary, and assessment. The introduction step includes a brief overview of the topic, new vocabulary, real-life applications, lesson main ideas, and a discussion question to spark the student’s thinking. The concept explanation step is where the teaching of the concept begins. During this step, key concepts are explained thoroughly to the student. Students will be presented with at least 3 questions with follow-up questions, ensuring critical thinking and

understanding of the concept. If the student answers incorrectly, the answer will not be immediately given, but instead, hints will be provided to the student to help guide them in the right direction. After concept explanation, the student will solve an example problem together with the avatar, in the example problem step. This step consists of a walk-through problem being solved in conversation style; the avatar will ask the student for their thoughts on what the next step for a problem may be, and ask critical thinking questions about the given problem. After completing some example problems, the student will complete practice problems that require them to apply the newly learned concept. These problems will come in a variety of types including multiple-choice, fill-in-the-blank, drag-and-drop, matching, and more. For each practice problem, if the student answers incorrectly, the avatar will not immediately provide the answer, but instead pinpoint the student’s trouble point and guide them in the right direction from there. After the completion of practice problems, the avatar will summarize key points that were covered during the lesson to reinforce learning. Lastly, as part of the mastery learning assessment model (MLAM), the student will undergo an assessment where the avatar will not be available for help. Results from the assessment will be used to determine mastery of a concept. This structure is set up to follow the MLAM based on other intelligent tutoring systems.

**3.3.2 Lesson Generation.** Lesson content for MathWiz uses BCPS’ standards and curriculum as a framework. However, the content is not regurgitated, each MathWiz lesson is generated by our lesson generation engine. The engine uses the predefined information as a base and tweaks the content, language, visuals, and examples to suit the user’s individual needs and culture. Collected user data is also leveraged to create culturally relevant content that is relatable to the user. For example, if a visual student who likes TV shows were to do the “Variables and Expressions” lesson, examples throughout the lesson would include some of their favorite shows. This student would also receive more visual explanations for concepts as opposed to a non-visual learner on the system. Additionally, each lesson includes skill-appropriate exercises that align with the user’s proficiency and mastery goals. Interactive elements such as a Desmos-powered graph, as shown in Fig. 4, are included to allow users to interact with the content. More interactive elements such as gamified activities will be added in the future to accommodate more lessons and users.

## 4 Conclusion

MathWiz is a revolutionary AI-powered intelligent tutoring system that addresses systemic inequities in mathematics education by integrating culturally relevant learning, mastery learning, and advanced technologies. With adaptive learning engines, dynamic lesson generation, personalization, and engaging 3D avatar companions, MathWiz delivers personalized and inclusive learning experiences tailored to diverse student needs. By incorporating gamification, culturally relevant content, and real-time interactivity, the platform empowers learners from underserved communities, bridging gaps in traditional education and fostering deeper engagement. Future work will involve a three-phase approach to enhance and evaluate MathWiz: (i) Co-design workshops with students and teachers to refine features and strategies, (ii) usability studies to

evaluate system interface and logic, and (iii) pilot testing to assess preliminary effectiveness. Participants will include 50–200 middle and high school students, math teachers, and parents from Baltimore City Public Schools. Data will be collected through pre- and post-assessments, user profiles, interactive lessons, and surveys. These efforts will further personalize learning, broaden content diversity, and strengthen MathWiz's ability to close educational gaps, fostering equity and academic success.

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